

Unicorn Finance workshop dataset guide

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Participant guide to Unicorn Finance's synthetic core-lending dataset. It explains the fictional business scenario, the purpose and grain of all eight tables, their relationships, and common analysis paths.

Where to find the data

All workshop tables are managed Delta tables in the default namespace:

```
hc_workshop.core_lending
```

Example:

```
SELECT *  
FROM hc_workshop.core_lending.loan_application  
LIMIT 10;
```

Important disclaimer

Unicorn Finance is fictional and this dataset is entirely synthetic. It was created for the Home Credit Philippines enablement workshop, but it is **not Home Credit production data** and does not reproduce Home Credit's products or physical core-system schema.

Names, national IDs, phone numbers, applications, stores, contracts, payments, and collection events are fabricated. Treat synthetic PII as sensitive during the governance exercises.

Business scenario

Unicorn Finance's POS installment product is its acquisition engine. A brand-subsidised 0% smartphone promotion increases application and contract volume. A minority of originations then show first-payment default concentrated in a small number of retail locations and sales associates. The top 20% of stores also account for roughly 80% of in-store applications.

The dataset is **not limited to that promotion**. It covers the broader core-lending lifecycle and includes POS installment, cash loan, Unicorn Flex revolving credit, and Unicorn Visa records across a two-year window. The promotion is one deliberately detectable cohort for investigation.

Participants can use the data to:

- Understand product and origination performance.
- Analyze approval and conversion rates.
- Calculate delinquency and first-payment default.

- Identify store and associate concentrations.
- Measure collection outcomes.
- Apply governance controls to customer and underwriting data.

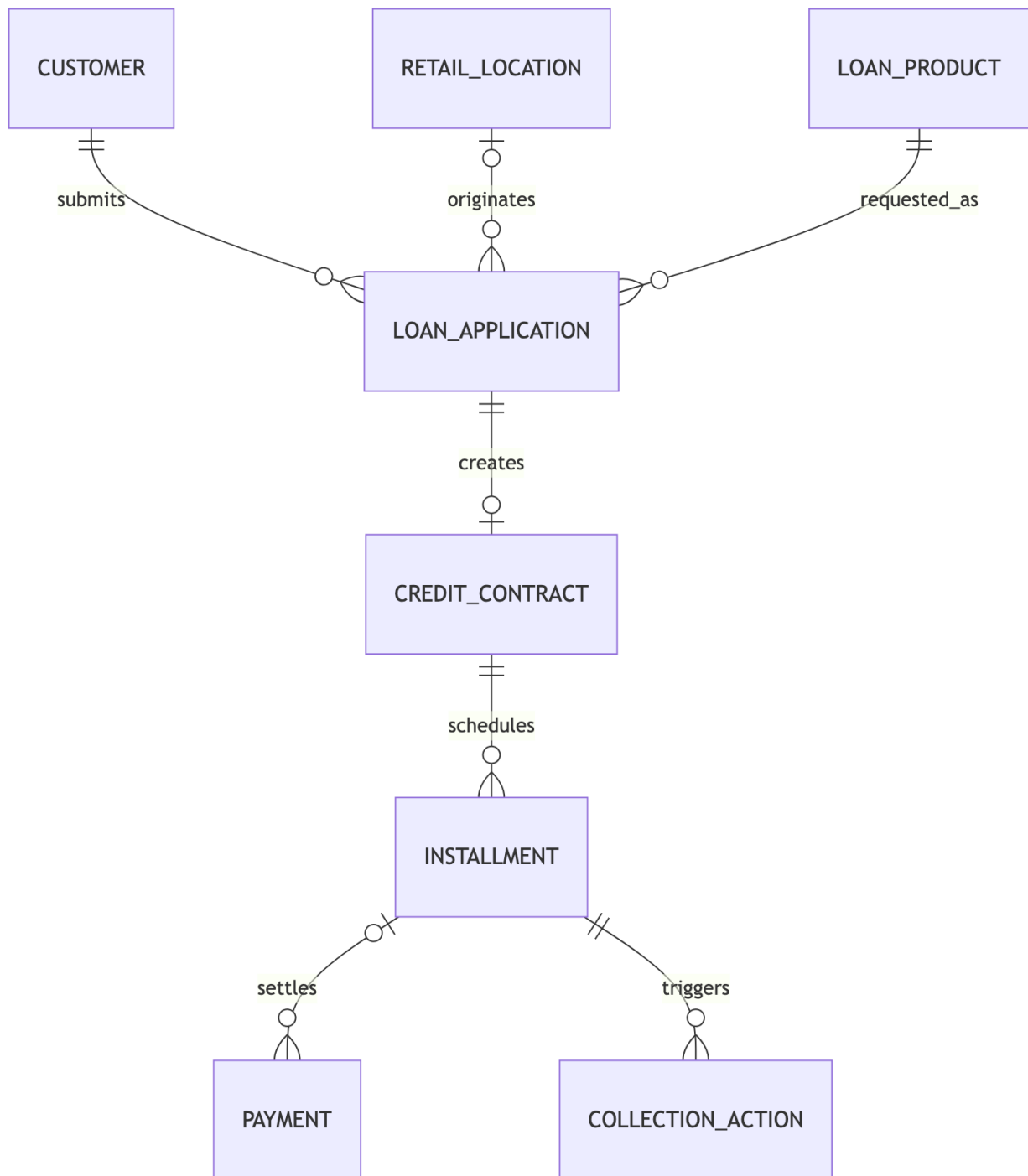
Schema at a glance

The schema follows the lending lifecycle from customer and product reference data through origination, servicing, and collections. **Grain** states exactly what one row represents.

Domain	Table	Grain	Purpose
Reference	customer	One borrower or prospect	Customer identity, contact, employment, income, and geography for origination and governance exercises
Reference	retail_location	One physical partner store	Merchant, store, and geographic context for volume and concentration analysis
Reference	loan_product	One product version	Commercial limits, tenor ranges, rates, and interest methods used to interpret applications and contracts
Origination	loan_application	One credit request	Requested product and amount, channel, store, associate, promotion, underwriting decision, and financed item
Origination	credit_contract	One approved application converted to a contract	Accepted financial terms, product structure, origination date, maturity, and lifecycle status
Servicing	installment	One scheduled fixed-term obligation	Due amounts, due date, settlement state, and as-of outstanding amount
Servicing	payment	One payment attempt	Posted receipts and failed attempts applied to an installment and contract
Collections	collection_action	One action against one delinquent installment	Collection method, DPD at action, outcome, promise to pay, and directly attributed payment

loan_application is the origination hub. Declined and cancelled applications do not produce contracts. Only fixed-term POS and cash contracts produce installments. Revolving products appear in applications and contracts but do not have generated transaction or statement tables in this workshop dataset.

Entity relationships



Relationship interpretation:

- One customer can submit many applications.
- An application references one product and optionally one retail location.
- An approved application creates at most one contract.
- A fixed-term contract has many installments.
- An installment can have multiple payment attempts with posted or failed status.
- A delinquent installment can trigger multiple collection actions.

To keep the diagram readable, it shows the primary analytical path. `payment.contract_id` also links every payment directly to its contract, and `collection_action.contract_id` links each collection action to the parent contract.

Primary join paths

Customer to application

```
FROM hc_workshop.core_lending.customer c
JOIN hc_workshop.core_lending.loan_application a
  ON c.customer_id = a.customer_id
```

Application to location and product

```
FROM hc_workshop.core_lending.loan_application a
LEFT JOIN hc_workshop.core_lending.retail_location l
  ON a.store_id = l.store_id
JOIN hc_workshop.core_lending.loan_product p
  ON a.product_id = p.product_id
```

`store_id` is nullable because digital and direct applications do not originate at a physical store.

Application to contract

```
FROM hc_workshop.core_lending.loan_application a
LEFT JOIN hc_workshop.core_lending.credit_contract c
  ON a.application_id = c.application_id
```

Use a left join when measuring application conversion because declined and cancelled applications have no contract.

Contract to installments and payments

```
FROM hc_workshop.core_lending.credit_contract c
JOIN hc_workshop.core_lending.installment i
  ON c.contract_id = i.contract_id
LEFT JOIN hc_workshop.core_lending.payment p
  ON i.installment_id = p.installment_id
```

One installment may have multiple payment attempts. Aggregate payments before joining when you need one row per installment.

Contract to collections

```
FROM hc_workshop.core_lending.credit_contract c
JOIN hc_workshop.core_lending.collection_action ca
  ON c.contract_id = ca.contract_id
```

Product concepts

- **POS installment** — Fixed-term purchase financing, usually originated in a partner store. Interest uses the add-on method unless a promotion reduces the customer rate to 0%.
- **Cash loan** — Unsecured cash financing, generally offered to existing customers with repayment history.
- **Unicorn Flex** — Revolving virtual credit line with a declining-balance interest method.
- **Unicorn Visa** — Revolving card product offered selectively to established customers.

Rates ending in `_rate_pct` are stored as percentage points. A value of `5.9900` means **5.99%**, not `0.0599`.

All monetary amounts are Philippine pesos unless the accompanying `currency_code` states otherwise.

Source fields versus derived metrics

The core tables contain business events and contractual facts. The following should be calculated during analysis rather than expected in the source:

- Current days past due and DPD bucket.
- First-payment-default flag.
- Origination vintage.
- Current contract balance.
- On-time payment ratio.
- Estimated interest and fee income.
- Credit loss and suspected-fraud indicators.

`days_past_due_at_action` is intentionally stored on `collection_action`: it is an immutable snapshot explaining why the action was taken.

`installment.outstanding_amount` and `installment_status_code` are servicing state as of the configured dataset date. `payment` and `collection_action` are event histories. Moving the as-of date forward can therefore change existing installment state while adding later events.

For this workshop, **first-payment default (FPD)** means the first installment reached five days past due before full settlement. A settlement exactly five days after the due date counts as FPD. Only contracts whose first installment has reached that observation point by the declared as-of date belong in the denominator:

```
WHERE installment_no = 1
AND date_add(due_date, 5) <= DATE'2026-09-01'
```

Suggested questions

1. Which products and provinces generate the most applications?
2. What is the approval rate by product and channel?
3. Did the 0% smartphone promotion increase originations?
4. Which stores and associates have unusually high first-payment default?
5. How does cash-loan delinquency compare with POS installment delinquency?
6. What proportion of installments are paid on time, partially, or late?
7. Which collection actions produce the highest promise-to-pay and payment-received amounts?
8. Which columns should be masked or access-controlled?

Data quality expectations

- Every application must reference an existing customer and product.
- Physical in-store applications should have a valid `store_id`.
- Only approved applications should create contracts.
- Contract terms must be within the selected product's limits.
- Every installment must reference an existing contract.
- Posted payments must have a posting timestamp.
- Failed payments should have a failure reason.
- A `PAYMENT_RECEIVED` collection outcome must reconcile to a posted payment for the same installment, timestamp, and amount.
- Collection actions should occur only for delinquent obligations and never after an attributed payment.

For column-level definitions, open `data-dictionary.pdf` in the participant materials folder.